

(Some) Solutions to *Exponential Mayhem*

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$$1 = e^0 \tag{1}$$

$$2 = \lceil \log(e^{e^1}) \rceil$$

$$3 = \lceil e^1 \rceil$$

$$4 = \lceil e^{\log(e^{e^1})} \rceil$$

$$5 = \lceil \log(e^{10^1}) \rceil$$

$$6 = \lceil e^{\log(e^4)} \rceil \tag{2}$$

$$7 = \lceil \log(e^{e^{e^1}}) \rceil$$

$$8 = \lceil e^2 \rceil \tag{3}$$

$$9 = \lceil e^{\log(e^5)} \rceil \tag{4}$$

$$10 = 10^1$$

$$11 = \lceil \ln(10^{\ln(10^2)}) \rceil \tag{5}$$

$$12 = \lceil \ln(10^5) \rceil \tag{6}$$

$$13 = \lceil 10^{\ln(3)} \rceil \tag{7}$$

$$14 = \lceil e^{\log(e^6)} \rceil \tag{8}$$

$$15 = \lceil \log(\log(e^{10^{e^1}})) \rceil$$

$$16 = \lceil e^{e^1} \rceil$$

Equation 1 can be substituted into any of the equations that have a 1 in the exponent; it is just neater to not have to write e^{e^0} for e . Additionally, equations 2 to 8 use numbers that are not equivalent to 0 or 1; it is quite clear that you can simply substitute the expression that equates to that number. For example, in equation 3, you can substitute $\lceil \log(e^{e^1}) \rceil$ for 2. This substitution will result in an expression that follows the constraints of the puzzle.